

merge sort

mi

Lec-39

Lec_40

Practice questions part2:

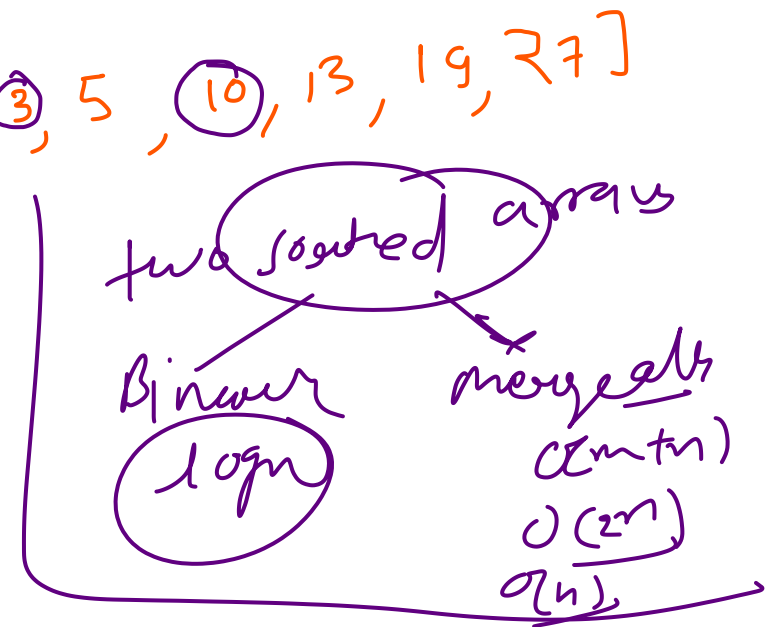
1.) 2 sorted arrays, A: size m, B: size n, TC for finding $A \cup B$ and $A \cap B$



$$1) A = [2, 3, 4, 10, 16], B = [1, 3, 5, 10, 13, 19, 27]$$

$$1) \underline{A \cup B} = [2, 3, 4, 10, 16, 1, 5, 13, 19, 27]$$

$$2) \underline{A \cap B} = [3, 10]$$



Approach 1:

$A = \{2, 3, 10, 15\}$
 m

$B = \{1, 3, 5, 10, 16, 19\}$
 n

$\Rightarrow O(n^2)$ ~~for A and B~~ for A and B
 $\text{for}(i=0; i < m; i++)$
{
 $\text{for}(j=0; j < n; j++)$
 {
 if ($a[i] == a[j]$)
 {
 $c[k] = a[i]$
 }
 }
}

~~for~~ $c = A \cap B$
array for A and B

$O(n^2)$

for common

$A \cup B \Rightarrow$ Same algorithm
 $\nwarrow$ count right once
 $O(n^2)$

✓

$A \cap B$

$n^2 \rightarrow n \log n \rightarrow n$
 $A = \{ \bullet, 0, 0, 0, 0, 7 \}$
 $\uparrow$ m 2 3 4

$B = \{ 0, 0, 0, 0, 0 \}$
 $\downarrow$ $2, 3, 5, 9$

Approach 2:

$n \log n$

for each element of ~~A~~ , apply Binary search on ~~B~~

$$\log n + \log n + \log n + \dots + m \log n$$

$$m = n$$

$n \log n$

$n \log n$

$ap = O(n^2)$
 $ap = O(n \log n)$

$\Rightarrow$ Simple two for loop
 $\Rightarrow$ or with Binary search

Approach 3:

merge algorithm

sorted

AND

$i=0, j=0$

while($i < m$ and $j < n$)

{
if ($a[i] == b[j]$)
{
print(a[i])
 $i++$, $j++$

if elements found common

else
{
if ($a[i] < b[j]$)
 $i++$

if element of a is smaller

else
{
 $j++$

if element of b is smaller

}

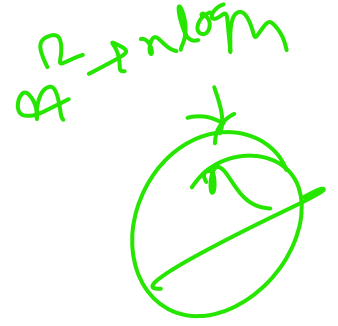
}

1) A n B

$A = \{1, 3, 5\}$
 $i \quad j \quad k$

$B = \{2, 3, 5, 10\}$
 $j \quad k \quad l \quad m$

[3, 5]



$O(m+n)$
if $m == n$
 $O(2n) \rightarrow O(n)$

2) AUB $\Rightarrow$

~~A = {1, 3, 5}~~ ~~B = {2, 3, 5, 10}~~
1, 3, 3, 5, 10

$\Downarrow$
remember copy
the remains
about

$O(m+n)$
 $O(m+n)$
 $O(n)$

$i=0, j=0$
while ($i < m$ and $j < n$)
{
 if ($a[i] == b[j]$)
 {
 print($a[i]$)
 $i++$ $j++$
 }
 else
 {
 if ($a[i] < b[j]$)
 {
 print($a[i]$)
 $i++$
 }
 else
 {
 print($b[j]$)
 $j++$
 }
 }
}

while ($i < m$)
{
 print($a[i]$)
 $i++$
}
while ($j < n$)
{
 print($b[j]$)
 $j++$
}

Result $\Rightarrow$ with the help of merge
algorithm $T_c \Rightarrow A \cup B$ and $A \cap B$

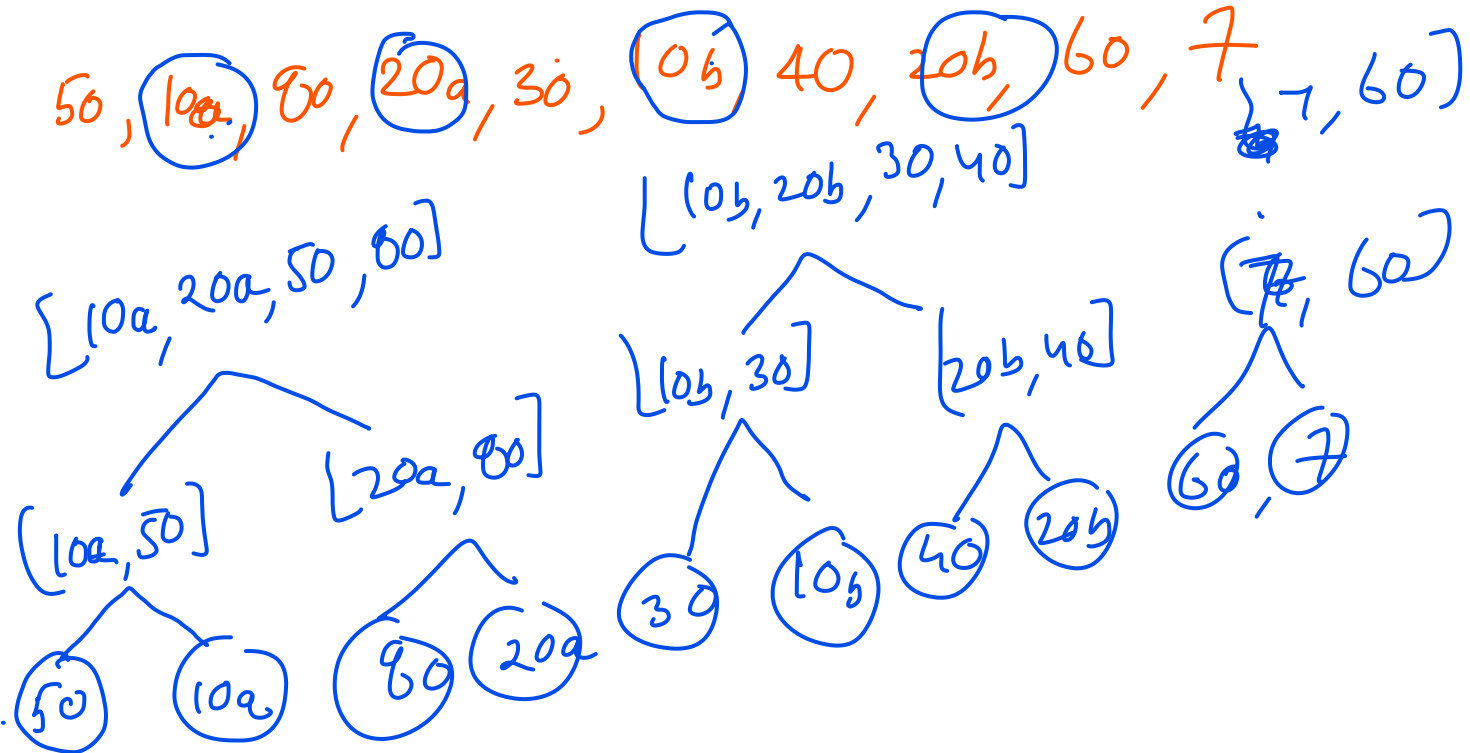
~~$\Rightarrow$~~ $O(n)$

2.) what is the o/p after second pass of merge sort?

Arr[] = 50, 10a, 80, 20a, 30, 10b, 40, 20b, 60, 7

Ans → 10a, 20a, 50, 80, 10b, 20b,
30, 40, 7, 60

Result after
 second pass
 o/p after 1st pass
 5 pass
 first pass



Stable sorting: if the position of elements remains same on

sorting is priority we can say that stable sorting

- merge sort is stable sorting
- merge sort is outplace $\Rightarrow O(n)$
- Tc of merge sort is $O(n \log n)$ each case.

Quick Soet

Thanks for

